

SMART CAMPUS AND PLACEMERNT USING AWS

Prepared in the partial fulfillment of the Summer Internship Program on AWS



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Thank you.

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ABSTRACT

The **Smart Campus Placement and Career Portal** is a cloud-based web application designed to simplify and improve the campus placement process by providing a centralized platform for students, recruiters, and placement administrators. The system enables students to register, create profiles, upload resumes, and apply for placement opportunities, while companies can post job openings and manage recruitment activities efficiently. The placement administrator monitors student registrations, company information, placement records, and recruitment processes through a secure administrative dashboard.

The application is deployed on **Amazon Web Services (AWS)** using cloud computing services to provide scalability, reliability, and secure access. The infrastructure consists of **Amazon EC2** for hosting the web application, **Amazon VPC** for network isolation, **Security Groups** for firewall protection, **Amazon S3** for storing resumes and placement-related documents, **Amazon CloudWatch** for monitoring system performance, and **Amazon Machine Images (AMI)** for creating reusable server instances. A relational database stores student details, company information, placement records, and administrator data to ensure efficient data management.

The portal offers various modules such as Home, Login, Student Registration, Company Management, Resume Management, Placement Details, Contact Us, and Admin Dashboard. These modules provide a user-friendly interface that reduces manual paperwork, minimizes errors, and enhances communication between students and recruiters.

By utilizing AWS cloud services, the proposed system provides improved performance, enhanced security, easy maintenance, high availability, and better resource utilization. The Smart Campus Placement and Career Portal serves as a modern cloud-based solution that streamlines placement activities and supports educational institutions in managing campus recruitment efficiently.

Keywords: AWS, Cloud Computing, EC2, Amazon S3, VPC, CloudWatch, HTML, CSS, Database, Placement Portal.

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CHAPTER 1

INTRODUCTION

Introduction

The rapid advancement of information technology and cloud computing has significantly transformed the way educational institutions manage academic and administrative activities. In today's competitive world, campus placements play a crucial role in shaping students' careers by providing employment opportunities before graduation. Every year, colleges organize campus recruitment drives where various companies visit institutions to recruit eligible students based on their academic performance, technical knowledge, communication skills, and aptitude. Managing these placement activities manually is a challenging and time-consuming process that often results in data redundancy, delayed communication, increased paperwork, and difficulty in maintaining accurate student records.

Traditionally, placement information such as student registration details, resumes, company eligibility criteria, interview schedules, and placement statistics are maintained using spreadsheets, paper documents, notice boards, and emails. These manual methods are inefficient, prone to human errors, and difficult to update regularly. Students frequently miss important placement notifications, while placement officers spend significant time organizing and verifying records. Companies also face challenges in collecting student information and filtering eligible candidates from large datasets. Therefore, there is a growing need for an intelligent, secure, and centralized digital platform that can efficiently manage the entire placement process.

The **Smart Campus Placement and Career Portal** is developed as a cloud-based web application to overcome these challenges by providing an integrated solution for students, placement officers, administrators, and recruiting companies. The system enables students to register online, maintain their academic profiles, upload resumes, and view placement opportunities through a user-friendly web interface. Companies can publish recruitment notifications, specify eligibility criteria, and review student applications, while placement administrators can manage student records, company details, placement drives, and recruitment reports through a secure administrative dashboard.

The primary objective of the proposed system is to automate campus placement management by replacing traditional manual processes with an efficient cloud-based solution. The portal ensures transparency, reduces paperwork, minimizes data duplication, and provides real-time access to placement-related information. Students can easily access company details, placement schedules, interview notifications, and recruitment updates from anywhere using an internet connection. This improves communication between students and placement officers while enhancing the overall efficiency of campus recruitment activities.

Cloud computing has emerged as one of the most influential technologies in modern software development because it provides scalable, reliable, and cost-effective computing resources over the internet. Instead of maintaining expensive physical servers, organizations can deploy applications on cloud platforms that offer better availability, automatic scalability, improved security, and simplified maintenance. Educational

institutions increasingly adopt cloud-based solutions because they reduce infrastructure costs while ensuring uninterrupted access to services.

To achieve these benefits, the Smart Campus Placement and Career Portal is deployed on **Amazon Web Services (AWS)**, one of the world's leading cloud computing platforms. AWS offers a comprehensive collection of cloud services that simplify application deployment, storage, networking, monitoring, and security. The application utilizes multiple AWS services to build a secure and reliable cloud environment capable of supporting placement management activities.

The web application is hosted using **Amazon Elastic Compute Cloud (EC2)**, which provides scalable virtual servers for running the portal continuously. **Amazon Virtual Private Cloud (VPC)** creates an isolated and secure virtual network where the application resources are deployed safely. **Security Groups** act as virtual firewalls that control incoming and outgoing network traffic to protect the application from unauthorized access. **Amazon Simple Storage Service (Amazon S3)** is used for storing resumes, placement documents, and other files securely with high durability and availability. **Amazon CloudWatch** continuously monitors the health and performance of the cloud infrastructure by collecting logs, system metrics, CPU utilization, memory usage, and network statistics. **Amazon Machine Images (AMI)** provide pre-configured server templates that simplify backup, recovery, and rapid deployment of additional application instances whenever required.

The portal consists of several functional modules including Home Page, Student Registration, Login, Resume Upload, Company Information, Placement Details, Contact Us, Admin Dashboard, and Logout. Each module performs a specific function that collectively contributes to the complete placement management process. The Home Page provides navigation and general information about the portal. The Registration module allows students to create new accounts by entering personal and academic details. The Login module authenticates users before granting access to system resources. The Resume module enables students to upload and manage their resumes securely. The Company module displays recruitment information, eligibility criteria, and available job openings. The Placement module provides placement statistics and selected student information. The Contact Us module offers institutional contact details for communication, while the Admin module enables placement officers to manage all records efficiently.

The database forms the backbone of the application by storing student profiles, company information, placement records, login credentials, and administrator details. A relational database management system is used to maintain structured and consistent data while ensuring efficient retrieval and secure storage of information. Proper database design minimizes redundancy, improves integrity, and supports future scalability of the application.

The proposed Smart Campus Placement and Career Portal provides numerous advantages over traditional placement management methods. It reduces manual effort, improves data accuracy, enhances communication, supports secure authentication, simplifies document management, and provides centralized access to placement-related information. Since the application is deployed on cloud infrastructure, it offers high availability, automatic scalability, disaster recovery capabilities, and efficient resource utilization. The use of

AWS services also strengthens application security through network isolation, monitoring, and controlled access mechanisms.

The system is designed to be user-friendly, reliable, scalable, and easy to maintain. It serves as an effective digital solution that simplifies campus recruitment activities while helping educational institutions improve placement management efficiency. Furthermore, the project demonstrates the practical implementation of cloud computing technologies in solving real-world educational challenges and provides students with valuable exposure to modern cloud platforms such as Amazon Web Services. The Smart Campus Placement and Career Portal therefore represents a significant step toward digital transformation in campus placement management, ensuring better coordination among students, placement officers, and recruiting organizations while supporting future enhancements and technological advancements.

OBJECTIVES

The primary objectives of the Smart Campus Placement and Career Portal are:

- To automate the campus placement process.
- To provide centralized management of placement information.
- To simplify student registration and profile management.
- To allow students to upload resumes securely.
- To maintain company and placement details.
- To provide secure login facilities for administrators and students.
- To deploy the application using Amazon Web Services.
- To monitor system performance using AWS CloudWatch.
- To improve communication between students and recruiters.
- To reduce manual paperwork and administrative effort.

SCOPE OF THE PROJECT:

The Smart Campus Placement and Career Portal provides an efficient platform for managing placement activities within educational institutions. Students can create profiles, upload resumes, and apply for placement opportunities through an online portal. Placement administrators can monitor student registrations, company details, and recruitment activities from a centralized dashboard.

The project utilizes cloud infrastructure to ensure scalability, security, and high availability. AWS services enable the application to be accessed from anywhere with an internet connection while maintaining reliable performance and secure data storage. The system can be extended in the future by integrating aptitude tests, interview scheduling, email notifications, analytics dashboards, and artificial intelligence-based job recommendations.

CHAPTER-2 LITERATURE SURVEY

Literature Survey

The rapid growth of information technology and cloud computing has significantly transformed the management of educational institutions. Among various academic activities, campus placement plays a vital role in helping students secure employment opportunities before completing their education. Traditionally, placement management has been carried out using manual methods such as paper records, spreadsheets, emails, and notice boards. Although these methods have been used for many years, they are time-consuming, prone to errors, and difficult to maintain as the number of students and recruiting companies increases.

With the advancement of web technologies, many educational institutions have started adopting web-based placement management systems. These systems provide online registration, placement notifications, student profile management, and company recruitment information through a centralized platform. Web-based applications improve communication between students, placement officers, and recruiters while reducing paperwork and administrative effort. However, many existing systems are deployed on local servers, making them difficult to scale and maintain. They often suffer from limited storage capacity, hardware failures, and reduced availability during high user traffic.

Cloud computing has emerged as an effective solution to overcome these limitations by providing scalable, reliable, and cost-effective infrastructure. Cloud platforms enable institutions to host applications on virtual servers, securely store large amounts of data, monitor system performance, and provide uninterrupted access to users from anywhere through the internet. Among various cloud service providers, Amazon Web Services (AWS) has become one of the most widely used cloud platforms due to its comprehensive set of services, global infrastructure, high availability, and strong security mechanisms.

Several research studies have focused on the implementation of cloud-based educational applications for student management, e-learning, and placement activities. These studies demonstrate that cloud computing improves system performance, reduces infrastructure costs, increases data security, and simplifies application deployment. Services such as Amazon EC2, Amazon S3, Amazon Virtual Private Cloud (VPC), CloudWatch, and Identity and Access Management (IAM) provide a secure and efficient environment for hosting web applications. These services allow institutions to build scalable solutions capable of handling thousands of users simultaneously while ensuring data integrity and availability.

Existing campus placement portals generally provide basic functionalities such as student registration, company registration, job notifications, and resume uploads. However, many systems lack centralized cloud deployment, secure network configuration, real-time monitoring, and efficient storage management. Some applications also do not provide dedicated administrator dashboards for managing placement records and monitoring recruitment activities effectively. These limitations reduce the overall efficiency and reliability of placement management systems.

The proposed **Smart Campus Placement and Career Portal** addresses these shortcomings by integrating modern cloud computing technologies with web application

development. The system is deployed on Amazon Web Services using Amazon EC2 for application hosting, Amazon VPC for secure network isolation, Security Groups for firewall protection, Amazon S3 for storing resumes and placement documents, Amazon CloudWatch for monitoring application performance, and Amazon Machine Images (AMI) for backup and rapid deployment. This cloud-based architecture ensures high availability, scalability, security, and simplified maintenance of the application.

The portal provides multiple modules including Home, Student Registration, Login, Admin Dashboard, Company Management, Resume Management, Placement Information, Contact Us, and Logout. These modules work together to automate the complete placement process while maintaining secure communication between users and the cloud infrastructure. The proposed system minimizes manual work, improves transparency, enhances communication between students and recruiters, and provides centralized management of placement-related information.

Overall, the literature survey indicates that cloud-based placement management systems offer significant advantages over conventional methods. By combining cloud infrastructure with modern web technologies, educational institutions can efficiently manage placement activities while ensuring better performance, reliability, scalability, and security. The proposed Smart Campus Placement and Career Portal builds upon these advancements to provide a comprehensive cloud-based solution for campus recruitment management.

Existing System

In the existing system, campus placement activities are primarily managed using manual or semi-automated methods. Student details, resumes, company information, and placement records are maintained through spreadsheets, physical documents, emails, or standalone desktop applications. Placement notifications are generally communicated through notice boards, messaging applications, or social media groups.

Although these methods are simple to implement, they introduce several operational challenges. Data redundancy, delayed communication, difficulty in maintaining records, and increased paperwork reduce the overall efficiency of placement management. Students may miss important notifications, while placement officers spend considerable time organizing and verifying placement data. Furthermore, existing systems often lack centralized storage, secure authentication, and proper monitoring mechanisms.

Limitations of Existing System

- Manual maintenance of student records.
- Large amount of paperwork.
- Difficulty in updating placement information.
- Data redundancy and inconsistency.
- Limited accessibility.
- Lack of centralized database management.
- Security vulnerabilities.
- Poor scalability.
- Absence of cloud infrastructure.
- No real-time monitoring.

Drawbacks of Existing System

The existing campus placement management system mainly relies on manual processes and traditional methods for maintaining student records, company information, and placement activities. Most institutions use spreadsheets, paper documents, emails, and notice boards to share placement-related information, which makes the entire process slow and inefficient. Managing a large volume of student data manually increases the possibility of errors, duplication, and loss of important information. Students often miss placement notifications due to delayed communication, while placement officers spend considerable time updating records and verifying student eligibility. The absence of a centralized platform makes it difficult to access placement information from different locations. Furthermore, traditional systems provide limited security, poor scalability, and inadequate data backup facilities. Since these systems are generally hosted on local infrastructure, they require regular maintenance and are vulnerable to hardware failures, resulting in reduced system availability and reliability. Overall, the existing system fails to provide a secure, efficient, and scalable solution for modern campus placement management.

Advantages:

The proposed Smart Campus Placement and Career Portal provides a modern, cloud-based solution for managing campus recruitment activities efficiently. The system offers a centralized platform where students, placement officers, and recruiters can access placement-related information from anywhere through the internet. By utilizing Amazon Web Services (AWS), the application ensures high availability, improved scalability, and enhanced security. Services such as Amazon EC2, Amazon VPC, Security Groups, Amazon S3, Amazon CloudWatch, and Amazon Machine Images (AMI) provide reliable application hosting, secure networking, efficient file storage, continuous system monitoring, and simplified backup management. The portal automates student registration, resume management, company information, placement records, and administrative tasks, significantly reducing manual effort and paperwork. It improves communication between students and recruiters through timely updates while maintaining accurate and secure records in a centralized database. The cloud-based architecture enables easy maintenance, efficient resource utilization, and future scalability, making the system suitable for educational institutions seeking a reliable and efficient placement management solution.

CHAPTER – 3 PROPOSED SYSTEM

Proposed System

- The proposed **Smart Campus Placement and Career Portal** is a cloud-based web application that automates campus placement management using Amazon Web Services (AWS). The system provides a centralized platform where students, placement administrators, and recruiters can perform placement-related activities efficiently.
- Students can register, log in, upload resumes, view company details, and apply for placement opportunities. Placement administrators can manage student information, company profiles, placement records, and recruitment activities through a secure administrative dashboard. The application is hosted on Amazon EC2, while Amazon S3 stores placement documents and resumes securely. Amazon VPC and Security Groups provide network security, CloudWatch continuously monitors server performance, and Amazon Machine Images (AMI) enable easy backup and deployment.
- The proposed solution reduces manual work, improves communication, enhances data security, and provides high availability through cloud infrastructure. It also offers scalability, allowing additional cloud resources to be allocated whenever the number of users increases.

Key Technologies Used:

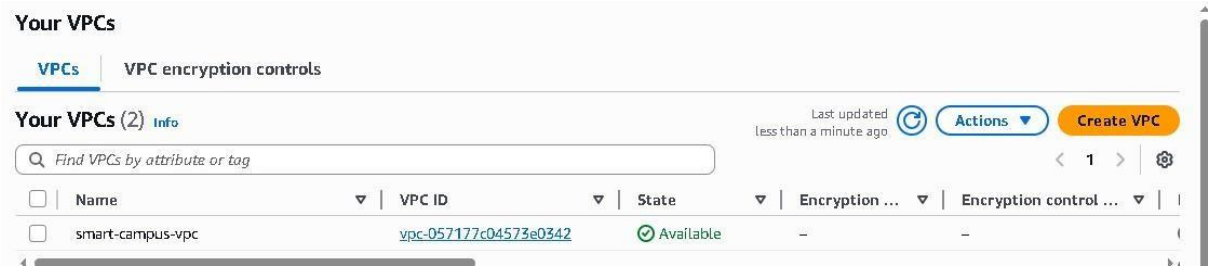
1. Amazon Web Services (AWS)

Amazon Web Services (AWS) is the cloud computing platform used to deploy and manage the Smart Campus Placement and Career Portal. AWS provides reliable, scalable, and secure cloud services that eliminate the need for maintaining physical servers. It enables the application to be accessed from anywhere through the internet while ensuring high availability, fault tolerance, and cost-effective resource utilization. The project uses various AWS services including Amazon EC2, Amazon VPC, Amazon S3, Security Groups, CloudWatch, and Amazon Machine Images (AMI) to build a complete cloud infrastructure.

2. Amazon Virtual Private Cloud (VPC)

Amazon Virtual Private Cloud (VPC) provides a secure and isolated virtual network for deploying AWS resources. It allows administrators to control IP address ranges, subnets, routing tables, and network gateways. By using VPC, the Smart Campus Placement and

Career Portal operates within a protected network environment where only authorized users can access application resources. This improves network security and enhances data privacy.



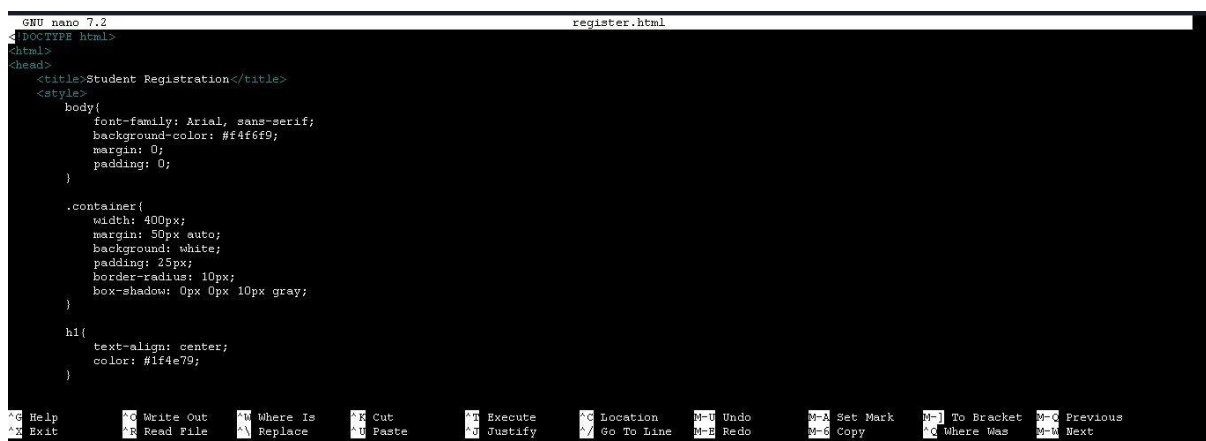
3. Security Groups

Security Groups act as virtual firewalls that control inbound and outbound traffic for the EC2 instance. They define rules that specify which users and services are allowed to access the application. In this project, Security Groups permit essential protocols such as HTTP, HTTPS, and SSH while blocking unauthorized traffic. This helps protect the server from external attacks and u



nauthorized access.

4. AMAZON EC2:



```
GNU nano 7.2 login.html
<!DOCTYPE html>
<html>
<head>
<title>Login - Smart Campus Placement Portal</title>
<style>
body{
font-family: Arial;
background:#f4f6f9;
text-align:center;
}
.login-box{
width:300px;
margin:100px auto;
background:white;
padding:20px;
border-radius:10px;
box-shadow:0px 0px 10px gray;
}
input{
width:90%;
padding:10px;
margin:10px;
}
button{
padding:10px 20px;
background:#1f4e79;

```

```
GNU nano 7.2 index.html
<!DOCTYPE html>
<html>
<head>
<title>Smart Campus Placement & Career Portal</title>
<style>
body{
font-family: Arial, sans-serif;
background-color:#f4f6f9;
margin:0;
padding:0;
text-align:center;
}

.header{
background:#1f4e79;
color:white;
padding:20px;
}

.header h1{
margin:0;
}

.nav{
margin-top:40px;
}

```

high performance and reliability. EC2 instances can be launched, stopped, restarted, or terminated whenever required, making application management simple and flexible. The Smart Campus Placement and Career Portal is deployed on an EC2 instance running Amazon Linux, which serves the application to users over the internet.

5. Amazon Simple Storage Service (Amazon S3)

Amazon S3 is an object storage service used to securely store resumes, placement documents, and other project files. It provides high durability, availability, and scalability for storing large amounts of data. The Smart Campus Placement and Career Portal stores placement-related files in an S3 bucket named **Smart Campus 2026**, enabling secure file management and easy retrieval whenever required.

Amazon S3 > Buckets > smart-campus-placement-2026

smart-campus-placement-2026 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [File systems - new](#) | [Access Points](#)

Objects (3) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	placement_details.txt	txt	June 30, 2026, 12:25:57 (UTC+05:30)	345.0 B	Standard
☐	sample_resume_1.txt	txt	June 30, 2026, 12:25:58 (UTC+05:30)	388.0 B	Standard
☐	sample_resume_2.txt	txt	June 30, 2026, 12:25:58 (UTC+05:30)	366.0 B	Standard

Buckets

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (1) [Info](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

☐	Name	AWS Region	Creation date
☐	smart-campus-placement-2026	Asia Pacific (Mumbai) ap-south-1	June 27, 2026, 12:41:34 (UTC+05:30)

Account snapshot [Info](#)

Updated daily

[View dashboard](#)

Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

RDS

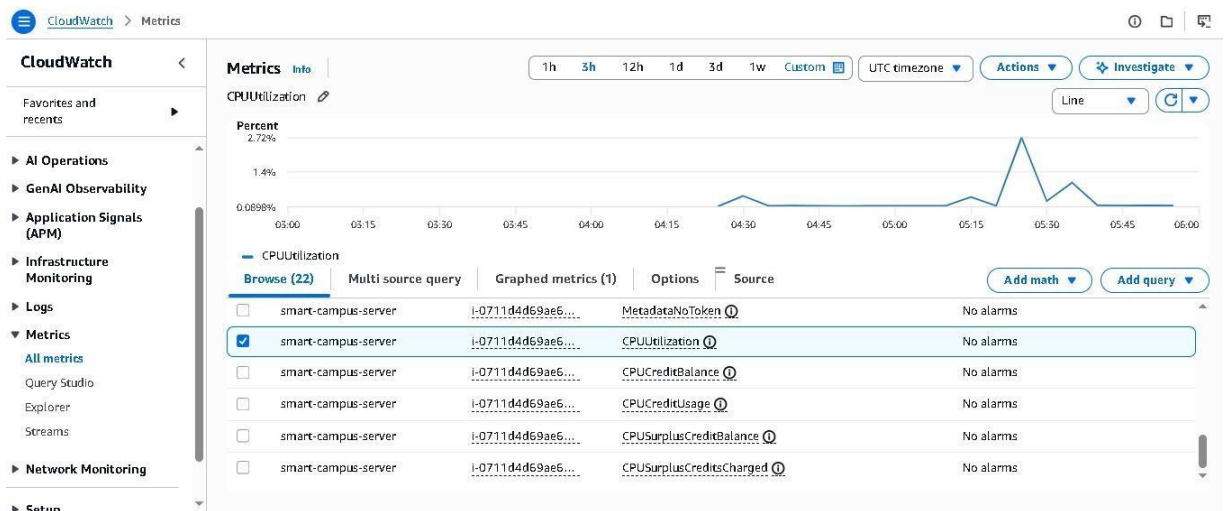
Databases (1) [Group resources](#) [Modify](#) [Actions](#) [Create database](#)

Filter by databases

☐	DB identifier	Status	Role	Engine	Upgrade rollout order	Region...	Size
☒	smart-campus-db	Availa...	Instance	MySQL C...	SECOND	ap-south-...	db.t4g.micro

6. Amazon CloudWatch

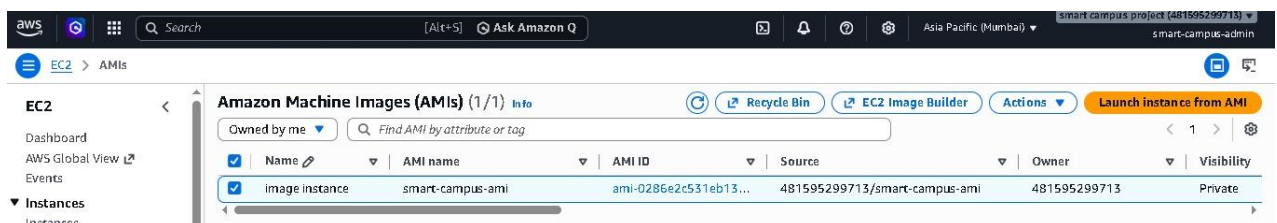
Amazon CloudWatch is a monitoring and logging service that continuously observes the performance of AWS resources. It collects important metrics such as CPU utilization, memory usage, disk activity, and network traffic. CloudWatch helps administrators



monitor application health, identify performance issues, and maintain the reliability of the placement portal through real-time monitoring and alerts.

7. Amazon Machine Image (AMI) :

Amazon Machine Image (AMI) provides a pre-configured template containing the operating system, application software, and configuration settings required to launch an EC2 instance. AMIs simplify deployment, backup, recovery, and replication of application servers. They enable rapid creation of identical virtual machines whenever additional servers are required.



8.HTML

HTML (HyperText Markup Language) is used to create the structure of the web pages in the Smart Campus Placement and Career Portal. It defines the layout of forms, buttons, menus, tables, images, and other user interface components. HTML serves as the foundation for developing responsive and interactive web pages.

9.CSS

CSS (Cascading Style Sheets) is used to enhance the appearance of the web application by applying colors, fonts, spacing, layouts, and responsive designs. CSS improves the visual presentation of the portal and provides a professional user interface that is easy to navigate.

10.Apache Web Server

Apache Web Server is used to host and deliver the web pages deployed on the Amazon EC2 instance. It receives HTTP requests from users and serves the appropriate HTML, CSS, and JavaScript files efficiently. Apache provides stable and secure web hosting for the Smart Campus Placement and Career Portal.

11.MySQL Database

MySQL is used as the relational database management system for storing student information, company details, administrator records, placement information, and login credentials. It provides secure, structured, and efficient data storage while supporting fast data retrieval and reliable database management.

Advantages of Proposed Work

The proposed **Smart Campus Placement and Career Portal** offers several advantages over the traditional placement management system by providing a secure, cloud-based, and centralized platform for managing campus recruitment activities. The system automates the complete placement process, reducing manual effort and improving operational efficiency. Students can register, upload resumes, access company information, and view placement opportunities through a single web application, while placement administrators can efficiently manage student records, company details, and placement activities.

One of the major advantages of the proposed system is its deployment on **Amazon Web Services (AWS)**, which provides a reliable and scalable cloud infrastructure. Services such as **Amazon EC2** ensure continuous application availability, **Amazon VPC** provides secure network isolation, **Security Groups** protect the application from unauthorized access, **Amazon S3** securely stores resumes and placement-related documents, **Amazon CloudWatch** continuously monitors system performance, and **Amazon Machine Images (AMI)** simplify backup and disaster recovery. These cloud services improve application reliability, security, and performance while reducing infrastructure maintenance.

The proposed system significantly enhances communication between students, recruiters, and placement officers by providing real-time access to placement notifications, company details, and recruitment information. Since all data is maintained in a centralized database, duplication and inconsistencies are minimized, resulting in improved data accuracy and integrity. The portal is accessible from anywhere through the internet, enabling users to perform placement-related activities without geographical limitations.

Another important advantage is the system's scalability and flexibility. As the number of students and companies increases, additional cloud resources can be allocated without affecting application performance. The cloud-based architecture also supports easy maintenance, automatic backups, and future enhancements. Overall, the Smart Campus Placement and Career Portal provides a secure, efficient, user-friendly, and cost-effective solution for educational institutions by simplifying campus placement management and improving the overall recruitment process.

Core Capabilities of the Proposed System

1. Student Registration

The system allows students to register by entering their personal, academic, and contact details through an online registration form. After successful registration, student information is securely stored in the database for future placement activities.

2. Secure User Login

The portal provides a secure login mechanism for students and administrators. Only authenticated users can access the system, ensuring the confidentiality and security of placement-related information.

3. Resume Management

Students can upload and manage their resumes through the portal. The resumes are securely stored in **Amazon S3**, allowing easy access and retrieval whenever required during the recruitment process.

1. Company Management

The system maintains detailed information about recruiting companies, including company name, job role, eligibility criteria, package details, and recruitment schedules. Students can view company information before applying for placements.

2. Placement Management

The placement module displays placement opportunities, selected student details, company-wise placement information, and recruitment updates. It helps students stay informed about ongoing and upcoming placement drives.

3. Administrator Dashboard

The administrator module enables placement officers to manage student records, company information, placement details, and uploaded resumes through a centralized dashboard. It simplifies the overall administration of campus recruitment activities.

4. Cloud-Based Deployment

The application is deployed on **Amazon EC2**, providing a reliable, scalable, and highly available cloud environment. Cloud deployment ensures uninterrupted service, better resource utilization, and simplified application management.

5. Secure Network Infrastructure

The proposed system uses **Amazon Virtual Private Cloud (VPC)** and **Security Groups** to create a secure network environment. These services protect the application from unauthorized access while allowing only permitted network traffic.

6. File Storage Management

The portal uses **Amazon S3** to store placement documents, resumes, and other important files. This provides secure, durable, and highly available cloud storage while eliminating the

limitations of local file storage.

7. Performance Monitoring

The system continuously monitors server performance using **Amazon CloudWatch**. It tracks CPU utilization, network activity, storage performance, and system health, enabling administrators to identify and resolve performance issues efficiently.

8. Backup and Recovery

The application utilizes **Amazon Machine Images (AMI)** to create backups of the EC2 instance. AMIs enable quick recovery of the application and simplify server replication whenever additional instances are required.

9. User-Friendly Web Interface

The Smart Campus Placement and Career Portal provides an intuitive and responsive web interface developed using HTML, CSS, and JavaScript. The simple navigation and interactive design enable students, administrators, and recruiters to perform placement-related activities efficiently with minimal effort.

CHAPTER -4

MODULES AND DATABASE DESIGN

Modules

1. Home Module

The Home Module serves as the entry point of the Smart Campus Placement and Career Portal. It provides an overview of the application along with navigation links to different sections such as Login, Student Registration, Company Details, Placement Information, Resume, Contact Us, and Logout. This module offers a user-friendly interface that enables users to access various services conveniently.

2. Student Registration Module

The Student Registration Module allows new students to create their accounts by entering personal, academic, and contact information. The entered details are validated before being stored in the database, ensuring that only valid information is maintained. This module forms the foundation of the placement management system by creating student profiles.

3. Login Module

The Login Module authenticates registered users before allowing access to the application. Students and administrators enter their login credentials, which are verified using the database. Successful authentication grants access to authorized modules, while invalid credentials are rejected to maintain system security.

4. Admin Module

The Admin Module enables placement administrators to manage the entire placement process through a centralized dashboard. Administrators can view student registrations, monitor placement records, manage company details, and supervise recruitment activities. This module simplifies administrative tasks and ensures efficient system management.

5. Company Module

The Company Module stores and displays information about recruiting companies, including company name, job roles, eligibility criteria, salary packages, and recruitment schedules. Students can access company information and stay updated about available placement opportunities.

6.Resume Module

The Resume Module allows students to upload, update, and manage their resumes. The uploaded resumes are securely stored in **Amazon S3**, ensuring safe storage, easy retrieval, and high availability. This module enables recruiters and administrators to access student resumes whenever required during the recruitment process.

7.Placement Module

The Placement Module manages placement-related information, including available job opportunities, company-wise placement details, selected students, and placement statistics. It provides students with updated placement information and helps placement officers monitor recruitment activities efficiently.

8.Contact Us Module

The Contact Us Module provides institutional contact information, including placement cell details, guide information, and official email addresses. Users can use this module to communicate with placement administrators regarding placement activities or technical issues related to the portal.

9 Logout Module

The Logout Module allows authenticated users to securely exit the application. It terminates the current user session and redirects the user to the Login page, preventing unauthorized access and ensuring application security.

These nine modules perfectly match your project screens:

- Home Page
- Login Page
- Student Registration Page
- Admin Page
- Company Page
- Resume Page
- Placement Page
- Contact Us Page
- Logout Page

DATABASE DESIGN

The database is one of the most important components of the **Smart Campus Placement and Career Portal**. It is responsible for storing, organizing, and managing all information related to students, administrators, companies, placement records, resumes, and user authentication. A well-designed relational database ensures data consistency, minimizes redundancy, and enables efficient retrieval of information. The proposed system uses a relational database to maintain secure and structured records while supporting various operations such as data insertion, updating, deletion, and retrieval.

The database design follows a normalized structure where each entity is maintained in a separate table with appropriate relationships. Primary keys uniquely identify each record, while foreign keys establish relationships between different tables. This approach improves database performance, maintains data integrity, and simplifies future enhancements.

The Smart Campus Placement and Career Portal consists of the following major database tables.

1. Students

The **Students** table stores the personal and academic information of registered students. It includes details such as Student ID, Name, Roll Number, Email ID, Department, Phone Number, Password, and Resume Path. This table acts as the primary source of student information and supports student authentication and placement activities.

2. Administrators

The **Administrators** table maintains the login credentials and profile details of placement administrators. It stores information such as Admin ID, Username, Email Address, Password, and Designation. The administrator manages students, companies, and placement records through the admin dashboard.

3. Companies

The **Companies** table contains information about organizations participating in campus recruitment. It includes Company ID, Company Name, Job Role, Eligibility Criteria, Salary Package, Recruitment Date, and Company Description. This table allows students to view company details and available placement opportunities.

4. Placements

The **Placements** table stores placement-related information, including Placement ID, Student ID, Company ID, Job Role, Package Offered, Selection Status, and Placement Date. This table helps maintain records of selected students and placement statistics.

5. Resumes :

The **Resumes** table stores information about resumes uploaded by students. It contains Resume ID, Student ID, Resume File Name, Upload Date, and Amazon S3 File Location. The actual resume files are securely stored in Amazon S3, while the database stores only the file references for easy retrieval.

6. Contact Information

The **Contact Information** table stores institutional contact details such as Department Name, Placement Officer Name, Email Address, Phone Number, and Office Address. This information is displayed on the Contact Us page of the application to facilitate communication between students and the placement cell.

Relationships Between Tables

The database tables are interconnected to maintain data consistency and efficient information retrieval. A one-to-many relationship exists between the **Students** table and the **Resumes** table, allowing one student to upload one or more resume versions. Similarly, a one-to-many relationship exists between the **Companies** table and the **Placements** table, as one company can recruit multiple students. The **Students** and **Placements** tables are also related, enabling the system to track placement details for each student. These relationships reduce data redundancy and maintain referential integrity throughout the database.

The database design provides a secure, scalable, and efficient foundation for the Smart Campus Placement and Career Portal. It supports fast data retrieval, reliable storage, secure authentication, and effective management of placement-related information, making the system suitable for academic institutions and future enhancements.

ER DIAGRAM

Entity Relationship (ER) Diagram Description

The Entity Relationship (ER) Diagram represents the logical structure of the **Smart Campus Placement and Career Portal** database. It illustrates the major entities of the system along with the relationships among them. The database is designed to store, organize, and manage student information, company details, resumes, placement records, administrator data, and contact information efficiently. A well-designed ER diagram minimizes data redundancy, maintains data integrity, and improves the overall performance of the application.

1. Students Entity

The **Students** entity stores the personal, academic, and login information of registered students. Each student is uniquely identified by a Student ID and contains details such as Name, Roll Number, Email, Department, Phone Number, Password, and Resume information. Students can register, log in, upload resumes, and participate in placement activities through the portal.

2. Administrator Entity

The **Administrator** entity manages the entire application. It stores administrator details such as Admin ID, Username, Email, Password, and Role. The administrator is responsible for managing student registrations, company information, placement records, and overall system operations.

3. Companies Entity

The **Companies** entity stores information about recruiting organizations. It includes attributes such as Company ID, Company Name, Job Role, Eligibility Criteria, Salary Package, Recruitment Date, and Company Description. Multiple students can apply to the same company during campus recruitment.

4. Resume Entity

The **Resume** entity maintains information about resumes uploaded by students. Each resume contains Resume ID, Student ID, File Name, Upload Date, and Amazon S3 Storage Path. The actual resume documents are securely stored in Amazon S3, while the database stores their references.

5. Placement Entity

The **Placement** entity stores information related to campus placements. It includes Placement ID, Student ID, Company ID, Job Role, Package Offered, Selection Status, and Placement Date. This entity maintains placement records of students selected by various companies.

6. Contact Entity

The **Contact** entity stores institutional contact details such as Placement Officer Name, Department, Email Address, Phone Number, and Office Address. This information is displayed on the Contact Us page for communication purposes.

Relationships Between Entities

1. A **Student** can upload one or more **Resumes**, establishing a **one-to-many (1:N)** relationship between Students and Resume.
2. A **Company** can recruit multiple **Students**, creating a **one-to-many (1:N)** relationship between Companies and Placement records.
3. A **Student** can have one placement record for each company, establishing a **one-to-many (1:N)** relationship between Students and Placements.
4. The **Administrator** manages Students, Companies, Placements, and Resume records, ensuring proper administration of the entire system.
5. The **Contact** entity remains independent and stores institutional communication details displayed within the application.

Overall ER Diagram Significance

The ER Diagram provides a structured representation of the database used in the Smart Campus Placement and Career Portal. It defines the relationships among various entities

while ensuring data consistency, normalization, and efficient data retrieval. The database design supports secure storage of student records, placement details, company information, and resumes while providing scalability for future enhancements. This logical database structure forms the foundation of the cloud-based placement management system and enables efficient interaction between all modules of the application.

6.5 Advantages of Database Design

A well-designed database is essential for the efficient operation of the **Smart Campus Placement and Career Portal**. The database stores all information related to students, administrators, companies, placement records, resumes, and contact details in a structured manner. Proper database design improves application performance, ensures data accuracy, and provides secure management of placement-related information. The following are the major advantages of the proposed database design.

1. Improved Data Consistency

The database is organized into separate tables with well-defined relationships, reducing data duplication and ensuring that all information remains accurate and consistent throughout the application.

2. Efficient Data Retrieval

The relational database enables fast retrieval of student records, company details, placement information, and resumes through optimized queries. This improves the overall response time of the application.

3. Enhanced Data Integrity

The use of primary keys, foreign keys, and database constraints ensures that only valid and meaningful data is stored. This prevents duplicate records and maintains strong relationships between different tables.

4. Secure Data Management

Sensitive information such as user credentials, student records, and placement details is stored securely in the database. Authentication mechanisms help prevent unauthorized access and protect confidential information.

5. Reduced Data Redundancy

Normalization techniques are used to eliminate duplicate data storage by organizing information into multiple related tables. This minimizes storage requirements and simplifies database maintenance.

6. Better Scalability

The database is designed to accommodate future growth in the number of students, companies, and placement records without affecting system performance. Additional tables and features can be incorporated easily whenever required.

7. Easy Maintenance

The structured database design simplifies database administration, allowing records to be inserted, updated, deleted, and retrieved efficiently. It also makes troubleshooting and future modifications easier.

8. Improved Application Performance

Optimized table structures and efficient relationships reduce query execution time, resulting in faster loading of student details, company information, placement records, and resume data.

9. Reliable Backup and Recovery

The database supports regular backup and recovery mechanisms, ensuring that important placement data can be restored in case of accidental deletion, hardware failure, or system malfunction. This improves the reliability of the overall application.

10. Supports Cloud-Based Storage

The database works efficiently with the AWS cloud infrastructure, while resumes and placement-related documents are securely stored in **Amazon S3**. This combination provides high availability, secure storage, and easy access to placement information from anywhere through the internet.

RELATIONSHIP BETWEEN ENTITIES

The **Smart Campus Placement and Career Portal** follows a relational database model in which all entities are interconnected to ensure efficient data management and maintain data integrity. The relationships between the entities help the system organize student information, company details, placement records, resumes, administrator activities, and contact information in a structured manner. Proper relationships eliminate data redundancy, improve consistency, and enable efficient retrieval of placement-related information.

1. Student – Resume Relationship

The **Student** entity is related to the **Resume** entity through the Student ID. Each registered student can upload and manage one or more resumes for different placement opportunities. Every resume belongs to only one student, establishing a **one-to-many (1:N)** relationship between the Student and Resume entities.

2. Student – Placement Relationship

The **Student** entity is associated with the **Placement** entity to maintain placement records. A student may participate in multiple recruitment drives and receive placement offers from different companies. Each placement record corresponds to a particular student, forming a **one-to-many (1:N)** relationship between Student and Placement.

3. Company – Placement Relationship

The **Company** entity maintains information about organizations participating in campus recruitment. A single company can recruit multiple students during placement drives. Therefore, one company is related to many placement records, establishing a **one-to-many (1:N)** relationship between Company and Placement.

4. Administrator – Student Relationship

The **Administrator** entity manages student registrations, profile updates, and placement activities. One administrator can manage the records of multiple students, making it a **one-to-many (1:N)** relationship between Administrator and Student.

5. Administrator – Company Relationship

The Administrator is responsible for adding, updating, and maintaining company details within the system. A single administrator can manage information for multiple companies, resulting in a **one-to-many (1:N)** relationship between Administrator and Company.

6. Administrator – Placement Relationship

The Administrator oversees placement drives, updates placement records, and verifies student selections. One administrator can manage multiple placement records, establishing a **one-to-many (1:N)** relationship between Administrator and Placement.

7. Contact – Administrator Relationship

The **Contact** entity stores institutional contact details such as the placement office information and official email addresses. These details are maintained by the administrator and displayed on the Contact Us page. This relationship ensures that users can communicate with the placement cell whenever required.

Overall Relationship

The relationships among the entities ensure smooth interaction between different modules of the Smart Campus Placement and Career Portal. Students register, upload resumes, and participate in placement drives, while administrators manage student records, company information, and placement activities. Companies conduct recruitment through the portal, and placement details are securely maintained in the database. These relationships provide a well-organized, reliable, and scalable database structure that supports efficient placement management and future system enhancements.

CHAPTER – 5

TESTING AND IMPLEMENTATION

System Test

System testing is a vital phase in the software development life cycle that verifies whether the complete application functions according to the specified requirements. It is performed after the successful completion of coding and integration testing to ensure that all modules of the system work together as a single integrated application. The primary objective of system testing is to identify errors, validate the functionality of the software, and confirm that the application meets both functional and non-functional requirements before deployment.

In the **Smart Campus Placement and Career Portal**, system testing was carried out to evaluate the performance, reliability, security, and usability of the application. Each module, including Home, Login, Student Registration, Admin Dashboard, Company Management, Resume Management, Placement Information, Contact Us, and Logout, was thoroughly tested using valid and invalid inputs. The testing process ensured that the application correctly processed user requests, stored information accurately in the database, and displayed appropriate outputs without errors.

Special attention was given to the cloud infrastructure deployed on **Amazon Web Services (AWS)**. The EC2 instance, Amazon S3 storage, Virtual Private Cloud (VPC), Security Groups, and CloudWatch monitoring services were tested to ensure proper connectivity, secure communication, file storage, and uninterrupted application availability. The database was also verified to ensure accurate storage, retrieval, updating, and deletion of records.

The results of system testing confirmed that all modules performed as expected. The application successfully handled user authentication, placement management, resume uploads, company information, and database operations while maintaining data integrity and security. The successful completion of system testing demonstrated that the Smart Campus Placement and Career Portal is reliable, secure, and suitable for practical deployment in educational institutions.

Types of Tests

Different testing techniques were performed to evaluate the functionality, performance, security, and reliability of the Smart Campus Placement and Career Portal. These testing methods ensured that each module operated correctly both individually and as part of the complete system.

Unit Testing

Unit testing is performed to verify the correctness of individual modules or components of the application. Each module, such as Login, Student Registration, Company Management, Resume Upload, and Placement Management, was tested independently to ensure that it produced the expected output for valid inputs. Unit testing helped identify coding errors during the initial stages of development and ensured that every module functioned correctly before integration.

Integration Testing

Integration testing verifies the interaction between different modules after unit testing has been completed. In this project, the communication between the Student Registration module, Login module, Database, Resume Management, Company Module, and Placement Module was tested. The objective was to ensure that data flowed correctly between modules and that all integrated components worked together without interface errors.

Functional Testing

Functional testing was performed to verify that every feature of the Smart Campus Placement and Career Portal operated according to the specified functional requirements. The testing included user registration, login authentication, company information display, resume upload, placement updates, administrator operations, and logout functionality. Both valid and invalid inputs were tested to confirm that the application generated appropriate responses and maintained data accuracy.

White Box Testing

White Box Testing focuses on examining the internal logic, program structure, and source code of the application. During development, the internal code, decision-making statements, loops, and validation logic were tested to ensure proper execution. This testing helped identify logical errors and improved the overall quality of the software.

Black Box Testing

Black Box Testing evaluates the functionality of the application without considering its internal implementation. Testers provided various input values and verified whether the application produced the expected outputs. This testing ensured that users could successfully perform registration, login, resume upload, company search, placement viewing, and logout operations through the graphical user interface.

Acceptance Testing

Acceptance Testing was conducted to verify that the Smart Campus Placement and Career Portal satisfied the project objectives and user requirements. The complete application was tested from the perspective of end users, including students and administrators. All major functionalities operated successfully, and the application met the expected performance, security, and usability standards. The successful completion of acceptance testing confirmed that the system was ready for deployment and practical implementation.

CHAPTER-6

RESULTS

The **Smart Campus Placement and Career Portal** was successfully designed, developed, and deployed using **Amazon Web Services (AWS)**. The implementation of the project demonstrated that cloud computing provides a reliable, scalable, and secure environment for managing campus placement activities. The application was hosted on an **Amazon EC2** instance within a secure **Virtual Private Cloud (VPC)**, while **Security Groups** were configured to control network access and ensure secure communication. Student resumes and placement-related files were stored in **Amazon S3**, and the overall performance of the application was monitored using **Amazon CloudWatch**. **Amazon Machine Images (AMI)** were used to simplify backup and recovery of the deployed application.

The developed portal successfully automated the major activities involved in campus placement management. Students were able to register, log in securely, upload resumes, view company information, and access placement details through a user-friendly web interface. The administrator efficiently managed student records, company details, placement information, and system operations through the administrative dashboard. The application performed all database operations such as insertion, updating, retrieval, and deletion of records accurately and efficiently.

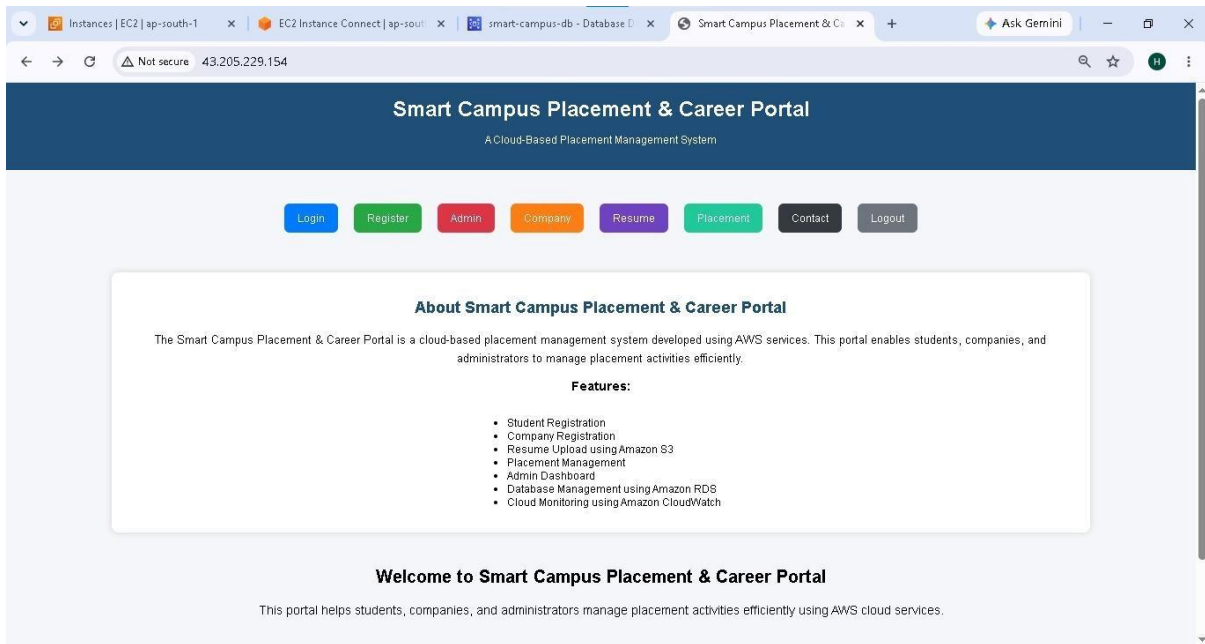
The web application was tested under different scenarios using valid and invalid inputs. All functional modules operated correctly without significant errors. The integration between the web application, database, and AWS cloud services functioned successfully, providing secure storage, reliable access, and continuous availability of the system. CloudWatch monitoring confirmed stable server performance throughout the testing process.

The output screens generated during project execution demonstrate the successful implementation of each module. The major output screens obtained from the developed system are presented below.

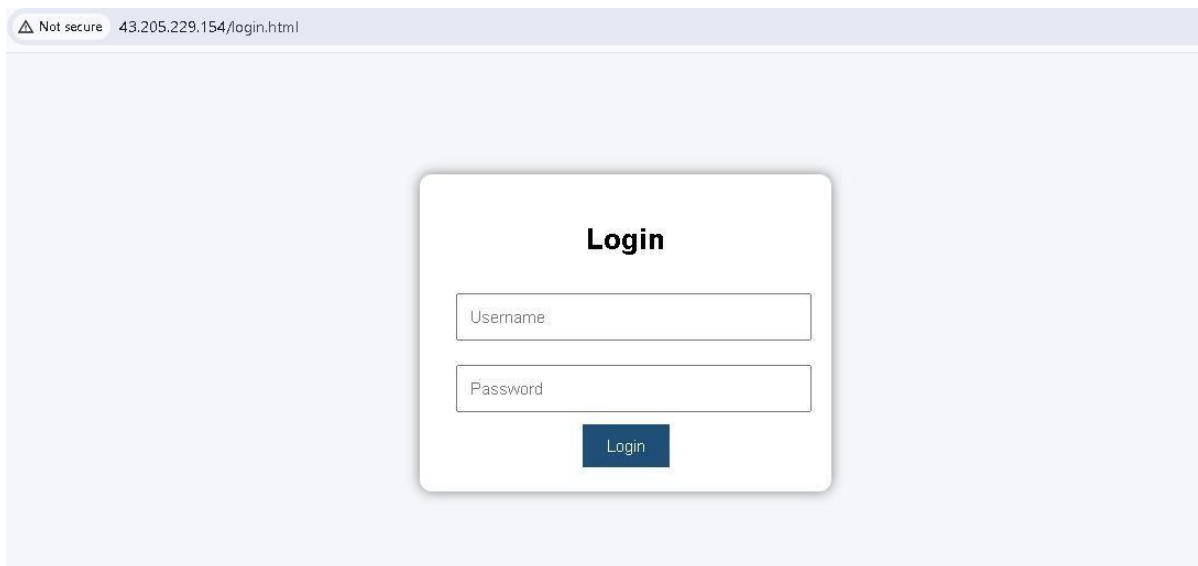
SCREESHOTS OF THE OUTPUT

Home Page :

The Home Page serves as the landing page of the application and provides navigation links to various modules including Login, Student Registration, Company Details, Resume, Placement Information, Contact Us, and Logout.

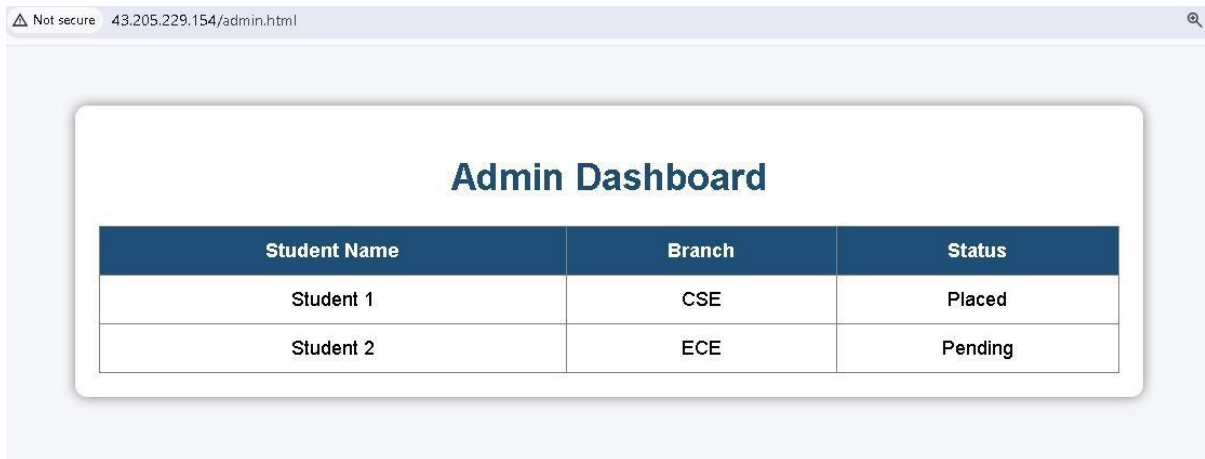


Login Page



The Login Page authenticates registered users by verifying their credentials before granting access to the application. It ensures secure user authentication and protects sensitive placement information.

Student Registration Page

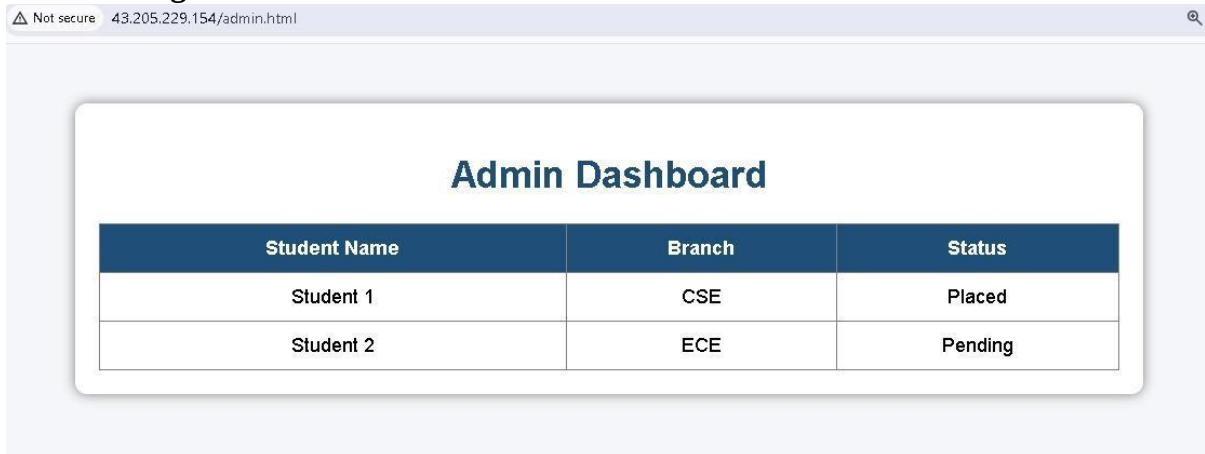


The screenshot shows a web browser window with the address bar displaying "43.205.229.154/admin.html". The page content is titled "Admin Dashboard" and features a table with the following data:

Student Name	Branch	Status
Student 1	CSE	Placed
Student 2	ECE	Pending

The Student Registration Page allows new students to create their accounts by entering personal, academic, and contact details. After successful validation, the information is stored securely in the database.

Admin Page



The screenshot shows a web browser window with the address bar displaying "43.205.229.154/admin.html". The page content is titled "Admin Dashboard" and features a table with the following data:

Student Name	Branch	Status
Student 1	CSE	Placed
Student 2	ECE	Pending

The Admin Page provides a centralized dashboard for placement administrators to manage students, companies, placement records, resumes, and other system activities efficiently.

Company Page

The Company Page displays detailed information about recruiting companies, including company name, job role, eligibility criteria, salary package, and recruitment schedule, enabling students to **view available placement opportunities**.

Not secure 43.205.229.154/company.html

Company Registration

Company Name

Email

Phone Number

Job Role

Package (LPA)

Register Company

Resume Page

Not secure 43.205.229.154/resume.html

Resume Upload

No file chosen

Upload Resume

The Resume Page enables students to upload and manage their resumes. The uploaded files are securely stored in Amazon S3, ensuring reliable storage and easy retrieval during the recruitment process.

Placement Page :

The Placement Page provides updated information about placement drives, selected students, company-wise recruitment details, and placement statistics, helping students stay informed about campus recruitment activities.

Placement Details			
Company	Role	Package	Eligible Branch
TCS	Software Engineer	4 LPA	ECE/CSE
Infosys	System Engineer	3.6 LPA	All Branches
Wipro	Project Engineer	4.5 LPA	ECE/CSE

Contact Us Page

The Contact Us Page displays the institutional contact information, including placement cell details, guide information, and official email addresses, enabling users to communicate with the placement authorities.

Login Page

Login	
	Username
	Password
Login	

The Logout Page securely terminates the user's session and redirects the user to the Login Page, preventing unauthorized access and ensuring the security of the application.

The successful implementation and execution of all the above modules confirm that the **Smart Campus Placement and Career Portal** effectively fulfills its intended objectives.

The results demonstrate that the proposed cloud-based solution provides an efficient, secure, scalable, and user-friendly platform for managing campus placement activities within educational institutions.

CHAPTER-7

CONCLUSION

The Smart Campus Placement and Career Portal has been successfully designed, developed, and implemented as a cloud-based web application to simplify and automate the campus placement process. The proposed system effectively addresses the limitations of traditional placement management by providing a centralized platform for students, placement administrators, and recruiting companies. It minimizes manual effort, reduces paperwork, improves communication, and enables efficient management of placement-related information through a user-friendly interface.

The project makes effective use of Amazon Web Services (AWS) to provide a secure, reliable, and scalable cloud environment. Services such as Amazon EC2, Amazon Virtual Private Cloud (VPC), Security Groups, Amazon S3, Amazon CloudWatch, and Amazon Machine Images (AMI) were successfully integrated to host the application, provide secure networking, store resumes and placement documents, monitor system performance, and ensure reliable backup and recovery. The use of cloud computing significantly enhances system availability, performance, security, and scalability while reducing infrastructure maintenance requirements.

The portal provides essential modules including Home, Student Registration, Login, Admin Dashboard, Company Management, Resume Management, Placement Information, Contact Us, and Logout, which together automate the complete placement management process. Students can easily register, upload resumes, access company information, and view placement opportunities, while administrators can efficiently manage student records, company details, and placement activities through a centralized dashboard.

The database design ensures secure and organized storage of student information, company records, placement details, and resumes while maintaining data integrity and minimizing redundancy. Comprehensive testing, including unit testing, integration testing, functional

testing, white box testing, black box testing, and acceptance testing, confirmed that all modules function correctly and meet the specified project requirements.

Overall, the Smart Campus Placement and Career Portal demonstrates how cloud computing can be effectively utilized to modernize campus recruitment activities. The proposed system provides an efficient, secure, reliable, and scalable solution that improves placement management, enhances user experience, and supports educational institutions in conducting campus recruitment more effectively. The project also serves as a practical example of implementing cloud technologies to solve real-world problems in higher education and provides a strong foundation for future enhancements and advanced placement management features.

9.2 Future Scope

The **Smart Campus Placement and Career Portal** has been developed to provide a secure, efficient, and cloud-based solution for managing campus placement activities. Although the current system successfully automates the placement process, there is significant scope for future enhancements to improve its functionality, performance, and user experience. With the continuous advancement of cloud computing, artificial intelligence, and web technologies, the system can be further expanded to support more advanced features and services.

One of the major future enhancements is the integration of **Artificial Intelligence (AI)** and **Machine Learning (ML)** algorithms to recommend suitable job opportunities for students based on their academic performance, technical skills, certifications, and interests. Intelligent recommendation systems can help students identify the most relevant companies and improve their chances of placement.

The application can also be enhanced by incorporating an **Online Aptitude Test and Technical Assessment Module**. This feature would allow students to participate in online examinations, coding tests, and technical assessments directly through the portal. The system can automatically evaluate results and shortlist eligible candidates based on predefined criteria.

Another valuable enhancement is the integration of **Interview Scheduling and Notification Services**. Automated scheduling of interviews, along with email and SMS notifications, would help students receive timely updates regarding recruitment drives, interview timings, and placement announcements. This feature would improve communication between students, placement officers, and recruiters.

The portal can further be extended by developing dedicated **mobile applications** for Android and iOS platforms. A mobile application would provide students with easy access to placement information, company updates, resume management, and notifications from anywhere at any time.

Future versions of the system may also include **advanced analytics and reporting dashboards** for placement administrators. Interactive reports and graphical representations of placement statistics, company-wise recruitment, department-wise performance, and student success rates would support better decision-making and institutional plan. The project can be integrated with professional networking platforms such as **LinkedIn** and online coding platforms to automatically import student profiles, certifications, and technical achievements. This integration would simplify profile creation and improve the recruitment process.

Additional cloud services such as **AWS Lambda**, **Amazon RDS**, **Amazon SNS**, and **AWS IAM** can be incorporated to improve application scalability, database management, notification services, and access control. These services would further strengthen the security, reliability, and performance of the system.

In conclusion, the Smart Campus Placement and Career Portal provides a strong foundation for a modern cloud-based placement management system. By incorporating emerging technologies and additional cloud services, the application can evolve into a comprehensive career management platform that supports students, educational institutions, and recruiting organizations more effectively while meeting future technological and industry requirements.

9.3 Applications

The **Smart Campus Placement and Career Portal** has a wide range of applications in educational institutions and recruitment organizations. It provides a centralized, cloud-based platform for managing placement activities efficiently while improving communication between students, placement officers, and recruiters. The system simplifies placement management, reduces manual effort, and enhances the overall recruitment process. The major applications of the proposed system are as follows.

1. Campus Placement Management

The portal is primarily used for managing campus placement activities in colleges and universities. It maintains student records, company details, placement schedules, and recruitment information in a centralized database.

2. Student Profile Management

Students can create and maintain their personal, academic, and professional profiles through the portal. They can update their information whenever required, ensuring that recruiters have access to the latest details.

3. Resume Management

The application allows students to upload and manage their resumes securely. The resumes are stored in **Amazon S3**, making them easily accessible to placement administrators and recruiters during the recruitment process.

4. Company Recruitment Management

Recruiting companies can publish job opportunities, eligibility criteria, salary packages, and recruitment schedules. This enables students to view available opportunities and apply for suitable positions.

5. Placement Record Management :

The system maintains complete records of placement activities, including selected students, company-wise recruitment details, placement statistics, and recruitment history. This information helps institutions analyze placement performance effectively.

6. Secure Cloud-Based Application

The portal is deployed on **Amazon Web Services (AWS)**, ensuring secure access, high availability, scalability, and reliable performance. Cloud deployment enables users to access the application from anywhere using an internet connection.

7. Educational Institution Management

The system assists placement cells in educational institutions by automating administrative tasks such as student registration, company management, resume verification, and placement monitoring, thereby reducing manual workload.

8. Centralized Data Management

All placement-related information is stored in a centralized database, allowing authorized users to retrieve, update, and manage records efficiently while maintaining data consistency and security.

9. Recruitment Monitoring

Placement administrators can monitor recruitment drives, track student participation, manage placement statistics, and generate reports for institutional analysis and decision-making.

10. Future Career Development Platform

The portal can be further expanded into a comprehensive career development platform by integrating online assessments, interview scheduling, certification management, job recommendations, and career guidance services, making it a complete solution for student career management.

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